Sole Temperature API

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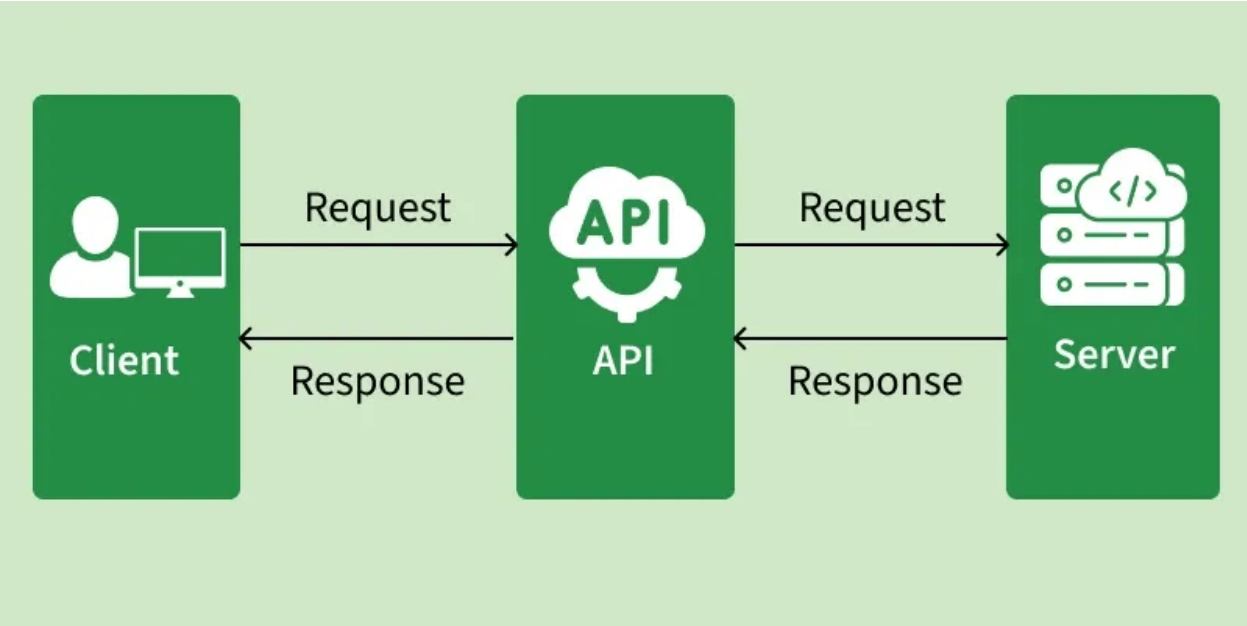
What is an API?

An API (Application Programming Interface) is a way for different software applications to communicate with each other.



Analogy:

1. You (client) request for order (data)
2. The waiter (API) takes the request to kitchen (Server)
3. The kitchen prepares the order
4. The waiter returns the response (the data requested)

Why use API?

APIs are important for building scalable, flexible, and connected programs. That is simple to deploy and maintain.

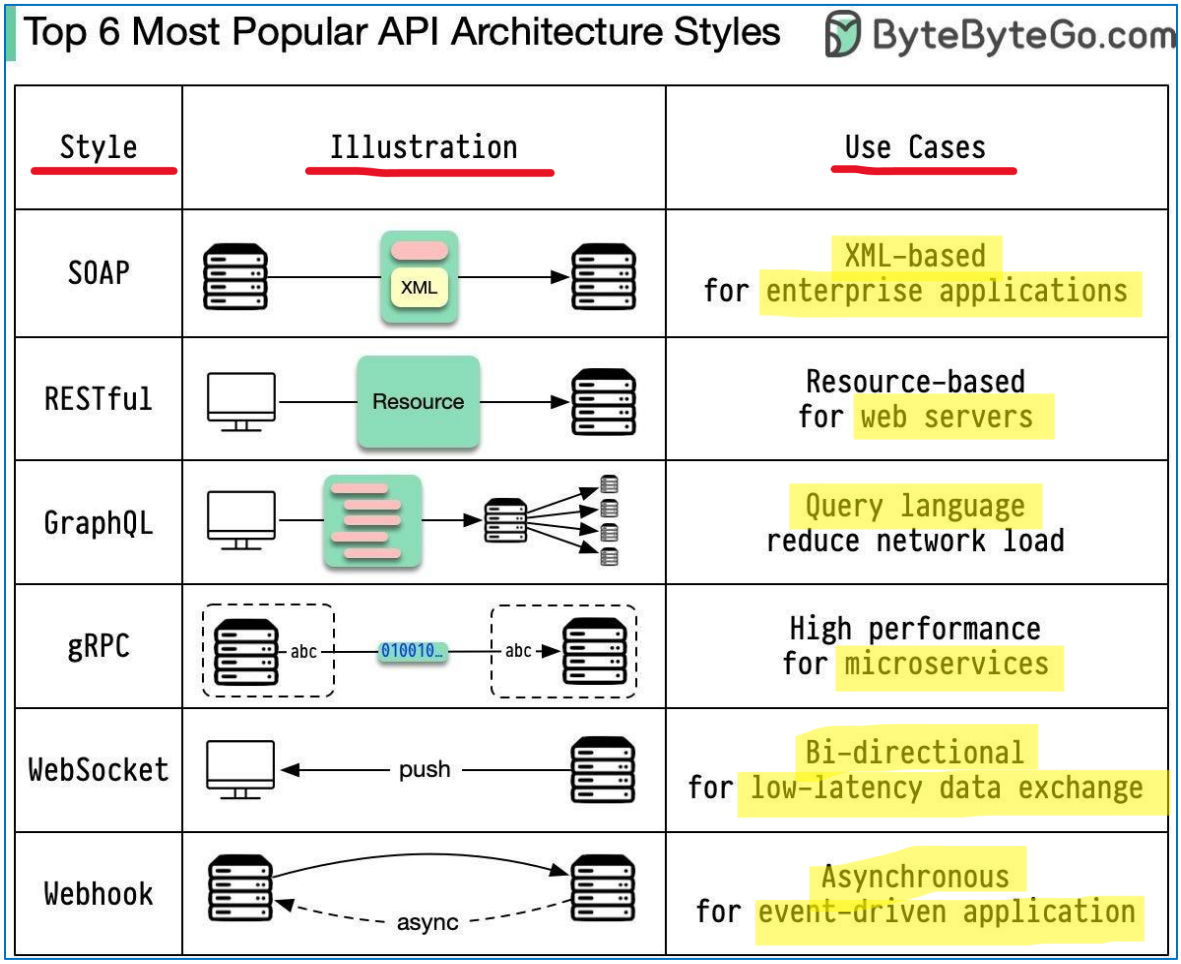
1. **Communication between system**: An API lets different software systems communicate with each other.
   Ex:
   Web/Mobile (Frontend) ---> Java (Backend) ---> MySQL (Database)

2. **Reusability**: APIs are reusable; they can serve multiple clients.
   Ex:
   - Mobile apps
   - Web app
   - Desktop app

3. **Secure**: API provides security in terms of indirect access.
   Ex:
   Client -> API -> Database
   The client does not have direct access to the database, preventing miss use.

Why do so many types of API exist?

Multiple API architecture exists to define how systems communicate and exchange data based on application needs.



REST - Most popular - uses HTTP request and returns a JSON format data. Common in Web-dev

SOAP - Old enterprise, but still common – use only XML, but has strong security. Common in banks/enterprise systems.

Sole Temperature API mechanics

The Sole Temperature uses a REST API. Specifically, it is calling the OpenWeather API, which is a REST API.
In this case, my Python program acts as a client, and the weather service is the API server.
Flow:

Python Script (client)

↓

HTTP Request

↓

OpenWeather REST API

↓

JSON Response

1. To request data from a web server, I use python library: **requests**
   - **Requests** allow you to send HTTP/1.1 requests extremely easily. There's no need to manually add query strings to your URLs, or to form-encode your POST data.
     Ex:
     r = requests.**get**('https://api.github.com/user')

2. Using the method: r**.status_code** . You can check HTTP status code such as:
   200(good), 404 (not found), 400 (bad request), etc...

3. Extract the data into JSON format:
   data = r.**json()**

```python

CityName = "Dallas"

API_KEY = os.getenv("OPEN_API_KEY")

URL = "https://api.openweathermap.org/data/2.5/weather"

CALL = f"{URL}?q={CityName}&appid={API_KEY}&units=imperial"

response = requests.get(CALL)

if response.status_code == 200:
    data = response.json()
```

NOTE: YOU'LL NEED AN API KEY, An API key is a unique secret code used to identify and authenticate an application when it accesses an API. It tells the server who is making the request. For: Security, Rate limits, Billing, etc... (Depends on the server service)

Security

... still working on it!...